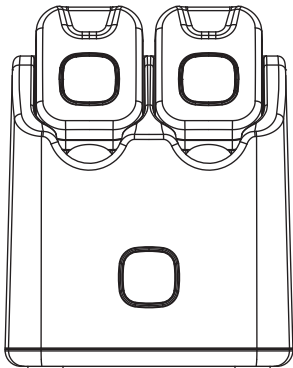


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# USER MANUAL

## SWS11

5.8GHz Wireless Audio Transmission System



This product is a high-performance 5.8GHz wireless audio transmission system comprising a transmitter, a receiver, and a dedicated charging case, with each unit operating in coordinated synergy. The transmitter handles audio capture and signal transmission, while the receiver manages wireless signal reception and audio output.

At its core, the system utilizes a dual-core floating-point DSP processor clocked at 320MHz. One core is dedicated to audio codec processing and system control to ensure a stable audio link, while the other is exclusively allocated for real-time AI inference. With physically isolated parallel processing, high-load AI computations do not compromise the real-time performance of audio codec operations.

The system delivers a typical end-to-end latency of 3ms and a transmission range of up to 50 meters in open environments. Utilizing multi-layer Adaptive Frequency Hopping (AFH) and a Neural Lossless Audio Engine (NLAE), it ensures stable, lossless audio transmission even in congested 5.8GHz bands, achieving a Total Harmonic Distortion plus Noise (THD+N) of <0.015%.

The transmitter operates at ≤90mA, and the receiver at ≤55mA. The devices provide over 5 hours of continuous operation and require approximately 2 hours for a full charge.

Supporting a one-to-many broadcast mode, the system allows up to 6 transmitters on different frequencies to operate simultaneously within the same environment, with a single transmitter supporting up to 32 receivers concurrently. This enables multiple receiving terminals to synchronously monitor the same audio source.

## 2、Wireless RF

### 2.1 Operating Frequency Band

Operating Frequency Band: 5.8GHz ISM (5725~5850MHz)

### 2.2 Multi-layer Adaptive Frequency Hopping (AFH)

This product utilizes multi-layer Adaptive Frequency Hopping (AFH) technology, integrating spectrum sensing, synchronous hopping, and closed-loop self-healing to establish a continuously self-optimizing

communication link within the congested 5.8GHz band.

- **Spectrum Scanning and Selection:** The system performs a full-band scan of the 5.8GHz spectrum to identify channel noise and occupancy. It automatically bypasses congested frequencies and locks onto the optimal channel combination for the current environment.
- **Cross-band Diversity Hopping:** The transmitter and receiver synchronously hop across multiple frequencies within a wide band at millisecond-level precision. Interference on a single frequency cannot suppress the entire link, as sporadic interference is absorbed through frequency diversity and error-correction coding.
- **Closed-loop Intelligent Frequency Switching:** The device continuously monitors channel link quality, rapidly switching to backup channels upon detecting interference. The switching process is seamless, ensuring zero audio interruption and remaining imperceptible to the user.
- **Link-loss Self-healing and Coexistence:** In extreme environments where link interruption occurs, the system automatically rotates through channel groups to rapidly re-establish the connection. By dividing the spectrum into multiple orthogonal channel groups with automatic allocation during pairing, the system supports the simultaneous, conflict-free coexistence of multiple device sets.

**The device supports manual frequency switching:** A short press of the transmitter's PAIR button cycles through 6 frequency color indicators (Red, Green, Blue, Yellow, Magenta, Cyan). The receiver automatically synchronizes the frequency and indicator color, further enhancing anti-interference capabilities in complex environments.

### 2.3 One-to-Many Transmission

This product supports broadcasting audio from a single transmitter to multiple receivers. All receivers maintain the same frequency hopping sequence and time synchronization, ensuring consistent audio output across all receiving terminals.

### 3.1 Sampling and Transmission

AD/DA Sampling: 96kHz / 24-bit

THD+N < 0.015%

Wireless Transmission Spec: 48kHz / 24-bit Lossless

**3.2 Frequency Response:** 20Hz ~ 20kHz,  $\pm 0.15\text{dB}$

**3.3 Dynamic Range:** 110dB

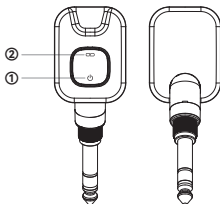
**3.4 End-to-End Latency:** 3ms (Typical)

**Note:** This metric is based on test results in an unobstructed, interference-free environment; slight variations may occur during actual use.

### 3.5 Neural-Network Lossless Audio Engine (NLAE)

Built on a dual-core floating-point DSP hardware architecture, the system features physically isolated parallel processing. One core handles audio codec and system control to ensure a stable audio link, while the other is dedicated to AI inference acceleration. AI computational loads do not compete for audio link resources or introduce additional latency.

1. **Neural Network Codec:** The AI model perceives audio content characteristics in real time, dynamically generating optimal encoding strategies for vocals, instruments, and silence, while intelligently allocating wireless bandwidth.
2. **Intelligent Predictive Repair:** During brief wireless signal loss, the receiver-side AI predicts and reconstructs the audio in real time based on preceding and subsequent audio features, maintaining audio continuity and integrity.
3. **Real-time Inference Acceleration:** AI inference and audio codec processing execute in parallel, delivering both lossless audio quality and low latency.



#### ① Power Button (ON/OFF)

- Long press: Power on or power off.
- Short press: Cycles through 4 input volume levels: -3dB, 0dB, 2dB, and 6dB (factory default is 0dB). During adjustment, the battery indicator will flash 5 times before returning to the standard battery display.

#### ② Frequency Switch Button (PAIR)

- Short press: Cycles through 6 frequency channels. The LED indicator will change color synchronously. Long press is invalid.
- Short press on the transmitter: Switches the frequency, and the receiver will follow suit. The pairing remains active; this function helps avoid environmental interference.
- Short press on the receiver: Switches the frequency without the transmitter following. The original pairing is cancelled, and the receiver will automatically pair with any transmitter operating on the same frequency.

#### ③ Charging Case

##### Power Button (ON/OFF)

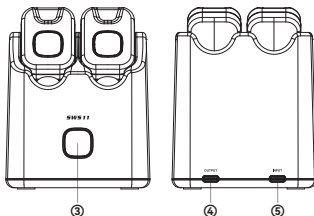
- Single click: Wake up / Power on. Double click: Power off.
- When powered off, inserting a wireless device into any slot will activate and power on the case.

#### ④ OUTPUT Power Port

- Provides external power supply to other devices.

## ⑤ INPUT Power Port

- Used to charge the charging case.



## 5、LED Indicator Status

EN

| Device Status                        | Transmitter LED                                                        | Receiver LED                                                                                                                       |
|--------------------------------------|------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Powered Off                          | Remembers Pre-shutdown Color                                           | Remembers Pre-shutdown Color                                                                                                       |
| Powered On                           | Displays Remembered Color                                              | Displays Remembered Color                                                                                                          |
| Connected                            | Solid On                                                               | Solid On                                                                                                                           |
| Disconnected                         | Flashing (Awaiting Connection)                                         | Flashing (Awaiting Connection)                                                                                                     |
| Short Press PAIR to Switch Frequency | Switch color; 6-color cycle; auto-switching when receiver is connected | Switch color; disconnect current transmitter when connected, then pair with the transmitter of the next color in the 6-color cycle |

**There are 6 frequency colors in total:** Red, Green, Blue, Yellow, Magenta, and Cyan.

### ■ Battery LED Indicator:

#### **Transmitter & Receiver (Four-corner Blue LEDs)**

- Powered On: Displays real-time battery level, with each segment representing 25%.
- Charging: LEDs flash sequentially. All 4 segments remain solid when fully charged. Wireless devices enter sleep mode and remain powered off.
- Fully Charged:

When the charging case INPUT is connected to a power source, all 4 blue LEDs on both the case and wireless devices remain solid. Wireless devices enter sleep mode and remain powered off.

When the charging case INPUT is disconnected, all 4 blue LEDs on both the case and wireless devices turn off. Wireless devices enter sleep mode and remain powered off.

- Low Battery: LEDs turn off, and the device automatically powers off for protection.

#### **Charging Case (Four-corner LEDs)**

- Single click to power on / Double click to power off: LEDs display real-time battery level, with each segment representing 25%.
- INPUT Connected for Charging: LEDs flash; LEDs remain solid when fully charged.
- OUTPUT External Power Supply: The charging case automatically powers on when an external device is connected. Wireless devices can be charged, and LEDs remain solid. Double-click the button to stop external power supply, or LEDs will turn off and the case will automatically power off when the battery is low.
- Charging Wireless Devices: LEDs remain solid. When the charging case INPUT is connected and wireless devices are fully charged, LEDs will remain on. When the charging case INPUT is disconnected and wireless devices are fully charged, LEDs will turn off and the case will automatically power off.
- Low Battery: LEDs turn off, and the case automatically powers off.

**Note:** Inserting wireless devices into the charging case activates charging mode with solid LEDs. Double-click the button to power off.

### ■ Charging Precautions:

1. Disconnecting the USB cable while the charging case INPUT is charging will cause the case to automatically power off.
2. Disconnecting the USB cable while the charging case OUTPUT is supplying external power will cause the case to automatically power off.
3. Please use the original charging cable to prevent excessive battery discharge.

| Item                             | Specifications                                                |
|----------------------------------|---------------------------------------------------------------|
| Battery Model                    | Charging Case: 505060, Receiver: 702025, Transmitter: 802525  |
| Battery Capacity                 | Charging Case: 2000mAh, Receiver: 300mAh, Transmitter: 500mAh |
| Nominal Battery Voltage          | 3.7V                                                          |
| Charging Case Single-Port Output | 5V <sup>⎓</sup> 1A                                            |
| Charging Case Total Output       | 5V <sup>⎓</sup> 1A                                            |
| Charging Case Input              | 5V <sup>⎓</sup> 2A                                            |
| Transmitter Input                | 5V <sup>⎓</sup> 500mA                                         |
| Receiver Input                   | 5V <sup>⎓</sup> 500mA                                         |
| Battery Life & Charging          | Operating Time > 5H; Charging Time = 2H                       |

- **Ultra-low Latency:** Typical end-to-end latency of 3ms.
- **Transmission Range:** Up to 50 meters in open environments.
- **Lossless Audio:** 96kHz/24-bit AD/DA sampling; 48kHz/24-bit lossless wireless transmission; 110dB dynamic range; 20Hz–20kHz frequency response ( $\pm 0.15$ dB).
- **Multi-layer Adaptive Frequency Hopping (AFH):** Features spectrum scanning and selection, cross-band diversity hopping, closed-loop intelligent frequency switching, and link self-healing to ensure a stable connection in complex 5.8GHz environments.
- **NLAE (Neural-Network Lossless Audio Engine):** AI-driven encoding strategies, predictive packet loss repair, and dual-core parallel inference acceleration.
- **One-to-Many Transmission:** A single transmitter can simultaneously connect to multiple receivers.
- **Manual Frequency Switching:** Manually select clean frequencies via the transmitter's PAIR button to enhance anti-interference capabilities.

## 8、Safety Precautions

- Do not use this device in high-temperature, humid, or flammable/explosive environments.
- Avoid severe impacts and exposure to water to prevent internal circuit damage.
- Keep children away during charging and ensure the environment is well-ventilated.

|                                              |                                                      |
|----------------------------------------------|------------------------------------------------------|
| AD/DA Sampling Rate                          | 96kHz / 24bit                                        |
| Wireless Transmission Spec                   | 48kHz / 24-bit Lossless                              |
| Wireless Operating Band                      | 5.8GHz ISM (5725~5850MHz)                            |
| Frequency Hopping Scheme                     | Multi-layer Adaptive Frequency Hopping (AFH)         |
| End-to-End Latency                           | 3ms (Typical)                                        |
| Dynamic Range                                | 110dB                                                |
| Frequency Response                           | 20Hz-20kHz, $\pm 0.15$ dB                            |
| Total Harmonic Distortion plus Noise (THD+N) | < 0.015%                                             |
| Effective Transmission Range                 | 50 meters (Open Environment)                         |
| Processor                                    | Dual-core Floating-point DSP @320MHz                 |
| Audio Engine                                 | Neural-Network Lossless Audio Engine (NLAE)          |
| Operating Current                            | Transmitter $\leq 90$ mA, Receiver $\leq 55$ mA      |
| Charging Case Dimensions                     | 82L $\times$ 47.5W $\times$ 89H (mm)                 |
| Transmitter & Receiver Dimensions            | 84.5L $\times$ 30.5W $\times$ 45H (mm)               |
| Weight                                       | Charging Case: 124g; Receiver: 25g; Transmitter: 28g |
| Total Package Weight                         | 275g                                                 |
| Standard Accessories                         | USB Charging Cable $\times$ 1                        |

**Note:** The operating distance is based on test results in an unobstructed environment; the effective transmission range may be reduced in the presence of external interference such as Wi-Fi.

★ Changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.

This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to Part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation. This equipment generates uses and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference to radio communications.

However, there is no guarantee that interference will not occur in a particular installation. If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures:

- Reorient or relocate the receiving antenna.
- Increase the separation between the equipment and receiver.
- Connect the equipment into an outlet on a circuit different from that to which the receiver is connected.
- Consult the dealer or an experienced radio/TV technician for help.

#### FCC Radiation Exposure Statement

The device has been evaluated to meet general RF exposure requirement. The device can be used in portable exposure condition without restriction.

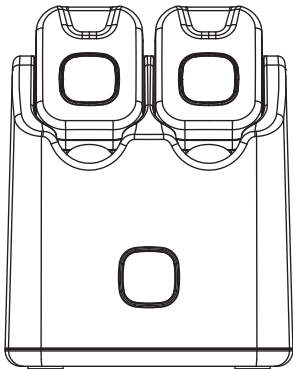
This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions:

- (1) This device may not cause harmful interference.
- (2) This device must accept any interference received, including interference that may cause undesired operation.

# 说明书

## SWS11

5.8GHz无线音频传输系统



本产品为高性能 5.8GHz 无线音频收发系统，由发射器、接收器及配套充电仓组成，各单元独立配套协同工作。发射器负责音频采集与信号发射，接收器完成无线信号接收以及音频输出。

系统核心采用主频 320MHz 双核浮点 DSP 处理器，一个内核承担音频编解码与系统主控，保障音频链路稳定；另一个内核专门用于 AI 实时推理。双内核物理隔离并行运行，AI 高负载运算不会影响音频编解码实时性能。

设备端到端典型延迟 3ms，传输距离可达 50米（空旷环境）；采用多层自适应跳频 AFH 技术以及神经网络无损音频引擎 NLAE，在干扰拥挤的5.8GHz 频段实现稳定无损音频传输，总谐波失真 THD+N < 0.015%。

发射器工作电流 $\leq 90\text{mA}$ ，接收器工作电流 $\leq 55\text{mA}$ ，设备连续工作时长 > 5 小时，充电时间约 2 小时。

系统支持一发多收广播模式，同一场景下，最多支持6个不同频点发射器同时使用，一个发射器可支持32个接收器同时使用。实现多接收终端同步监听同一音源。

### 2.1 工作频段

工作频段：5.8GHz ISM（5725~5850MHz）

### 2.2 多层自适应跳频（AFH, Adaptive Frequency Hopping）

本产品采用多层自适应跳频（AFH）技术，集频谱感知、同步跳变与闭环自愈于一体，在拥挤的 5.8GHz 频段构建持续自我优化的通信链路。

- **频谱勘测择优：**系统对 5.8GHz 频段进行全带扫描，识别各信道噪声与占用状态，自动避开拥堵频点，锁定当前环境品质最优信道组合。
- **跨频段分集跳频：**收发端在宽频区多个频点毫秒级同步跳变，单一频点干扰无法压制整条链路，偶发干扰被跳频分集与纠错编码吸收。
- **闭环智能换频：**设备持续监测信道链路质量，识别受干扰频点后快速切换备用信道；切换全程无缝，音频零中断，用户无感知。
- **断链自愈共存：**极端环境链路中断，系统自动轮换信道组快速重建连接；系统划分多组正交信道，配对自动分配信道，支持多套设备同场共存互不冲突。

**设备支持手动频点切换：**短按发射器 PAIR 配对键，可循环切换 6 种频点颜色（红、绿、蓝、黄、品红、青），接收器自动同步频点与指示灯颜色，进一步提升复杂环境抗干扰能力。

### 2.3 一发多收

本产品支持单发射器向多台接收器广播音频。全部接收器保持同一跳频序列与时间同步，保障各接收端音频输出一致性。

### 3.1 采样与传输

AD/DA 采样: 96kHz / 24bit

THD+N < 0.015%

无线传输规格: 48kHz / 24bit 无损

**3.2 频率响应:** 20Hz ~ 20kHz,  $\pm 0.15\text{dB}$

**3.3 动态范围:** 110dB

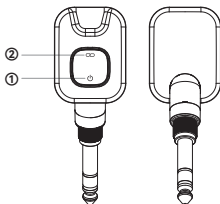
**3.4 端到端延迟:** 3ms (典型值)

注: 该指标为无遮挡、无干扰环境测试结果, 实际使用存在小幅波动。

### 3.5 神经网络无损音频引擎 (NLAE, Neural-Network Lossless Audio Engine)

依托双核浮点 DSP 硬件架构, 双内核物理隔离并行运行; 一核负责音频编解码、系统主控, 保障音频链路稳定; 另一核专职 AI 推理加速。AI 计算负载不会挤占音频链路资源, 不引入额外时延。

1. 神经网络编解码: AI 模型实时感知音频内容特征, 针对人声、器乐、静音不同场景动态生成最优编码策略, 智能分配无线带宽。
2. 智能预测修复: 无线信号短时丢失时, 接收端 AI 根据前后音频特征实时预测重建, 维持音频连贯完整。
3. 实时推理加速: AI 推理与音频编解码并行执行, 兼顾无损音质与低时延。

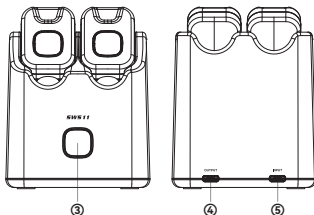


### ① 开关机键 (ON/OFF)

- 长按：开机或关机。
- 短按发射电源按钮，分别切换4档输入音量，对应-3db、0db、2db、6db，出厂默认0dB。调节时，电量灯为闪烁状态，闪烁5次后切换回电量显示。

### ② 频点切换按键 (PAIR)

- 短按：切换频点，6种频点颜色循环显示，LED同步变化，长按无效。
- 短按发射器，切换频点，接收器跟随切换，配对持续有效，此功能可避开环境中干扰。
- 短按接收器，切换频点，发射器不跟随，原配对取消，接收器可随即配对于与之频点相同的接收器。



### ③ 充电仓

#### 开关机键 (ON/OFF)

- 单击按钮开机激活 / 双击关机
- 关机状态下，无线设备插入任一孔位激活开机。

### ④ OUTPUT电源输出接口

- 对外给其他设备供电。

### ⑤ INPUT 电源输入接口

- 为充电仓本体充电。

## 五、LED 指示灯状态说明

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| 设备状态         | 发射器LED                 | 接收器LED                                |
|--------------|------------------------|---------------------------------------|
| 关机           | 记忆关机前颜色                | 记忆关机前颜色                               |
| 开机           | 显示记忆颜色                 | 显示记忆颜色                                |
| 正常连接         | 常亮                     | 常亮                                    |
| 未连接          | 闪烁等待连接                 | 闪烁等待连接                                |
| 短按 PAIR 切换频点 | 切换颜色，6色循环，接收器连接情况下跟随切换 | 切换颜色，发射器连接情况下断开原连接，将连接下一频点颜色的发射器，6色循环 |

频点颜色共 6 种：红、绿、蓝、黄、品红、青。

### ■ 电量LED显示:

#### 发射器、接收器（四角蓝色 LED）

- 开机状态：显示实时电量，每格表示 25% 电量。
- 充电状态：LED 循环闪烁，充满4格常亮，无线设备休眠不开机。
- 充满：

当充电仓INPUT接入时，充电仓及无线设备4格蓝色LED 常亮，无线设备休眠不开机。

当充电仓INPUT未接入时，充电仓及无线设备4格蓝色LED 熄灭，无线设备休眠不开机。

- 低电量：LED熄灭、设备自动关机保护。

#### 充电仓（四角 LED）

- 按钮单击开机/双击关机：LED显示实时电量，每格表示 25% 电量。
- INPUT 接入电源充电：LED 闪烁；充满后常亮。
- OUTPUT 对外供电：对外充电连接后，充电仓自动开机，无线设备可充电，LED 常亮；双击按钮停止对外供电，或电量不足，LED 熄灭自动关机。
- 充电仓为无线设备充电：LED 常亮；当充电仓INPUT接入时、无线设备充满，LED 不会熄灭。当充电仓INPUT未接入时、无线设备充满，LED 熄灭自动关机
- 低电量：LED 熄灭自动关机。

**提示：**无线设备插入充电仓，LED常亮，激活充电模式，可双击按钮关机。

### ■ 充电注意事项:

1. 充电仓 INPUT 充电状态下，拔除 USB 电源线，充电仓自动关机。
2. 充电仓 OUTPUT 对外供电时，拔除 USB 电源线，充电仓自动关机。
3. 请使用原装充电线缆，防止电池过度放电。

| 项目      | 参数                                  |
|---------|-------------------------------------|
| 电池型号    | 充电仓：505060, 接收器：702025, 发射器：802525  |
| 电池容量    | 充电仓：2000mAh, 接收器：300mAh, 发射器：500mAh |
| 电池标称电压  | 3.7V                                |
| 充电仓单口输出 | 5V $\overline{\text{---}}$ 1A       |
| 充电仓总输出  | 5V $\overline{\text{---}}$ 1A       |
| 充电仓输入   | 5V $\overline{\text{---}}$ 2A       |
| 发射器输入   | 5V $\overline{\text{---}}$ 500mA    |
| 接收器输入   | 5V $\overline{\text{---}}$ 500mA    |
| 续航充电    | 工作时长 > 5H; 充电时间=2H                  |

## 七、产品特性

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- **超低时延**：端到端典型延迟=3ms。
- **传输距离**：50米（空旷环境）。
- **无损音频**：96kHz/24bit AD/DA 采样；48kHz/24bit 无损无线传输；动态范围 110dB；频响 20Hz ~ 20kHz  $\pm$ 0.15dB。
- **多层自适应跳频 AFH**：频谱勘测择优、跨频段分集跳频、闭环智能换频、链路自愈，复杂 5.8GHz 环境保障链路稳定。
- **NLAE 神经网络无损音频引擎**：AI 智能编码策略、丢包预测修复、双核并行推理加速。
- **一发多收**：单发射器可同时连接多台接收器。
- **手动频点切换**：发射器 PAIR 按键手动切换干净频点，强化抗干扰能力。

## 八、安全注意事项

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- 禁止在高温、潮湿、易燃易爆环境使用本设备。
- 避免强力摔落、进水，防止内部电路损坏。
- 充电过程远离儿童，保证环境通风。

## 九、技术参数

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|               |                                                 |
|---------------|-------------------------------------------------|
| AD/DA 采样率     | 96kHz / 24bit                                   |
| 无线传输规格        | 48kHz / 24bit 无损                                |
| 无线工作频段        | 5.8GHz ISM (5725~5850MHz)                       |
| 跳频方案          | 多层自适应跳频 (AFH)                                   |
| 端到端延迟         | 3ms (典型值)                                       |
| 动态范围          | 110dB                                           |
| 频率响应          | 20Hz ~ 20kHz, $\pm 0.15\text{dB}$               |
| 总谐波失真 (THD+N) | < 0.015%                                        |
| 有效传输距离        | 50米 (空旷环境)                                      |
| 处理器           | 双核浮点 DSP @320MHz                                |
| 音频引擎          | 神经网络无损音频引擎 (NLAE)                               |
| 工作电流          | 发射器 $\leq 90\text{mA}$ , 接收器 $\leq 55\text{mA}$ |
| 充电仓尺寸         | 82L $\times$ 47.5W $\times$ 89H (mm)            |
| 发射器 & 接收器尺寸   | 84.5L $\times$ 30.5W $\times$ 45H (mm)          |
| 重量            | 充电仓: 124g; 接收器: 25g; 发射器: 28g                   |
| 包装总重          | 275g                                            |
| 标配附件          | USB 充电线 $\times 1$                              |
| 修订日期          | 2026-8-31                                       |

**备注：**工作距离为无遮挡环境测试结果；WiFi 等外部干扰存在时，有效传输距离会衰减。